AutoTerm-SST: Adaptive Terminology Memory with Zero Session-Time Setup for Streaming Speech Translation

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Abstract

We present AutoTerm-SST, an interactive system for terminology-aware streaming speech translation. Technical talks contain rare terms, but asking users to prepare a glossary or select a domain before every session creates substantial friction. AutoTerm-SST instead organizes a large offline terminology catalog into compact topic slices. A training-free router matches recent streaming translation context to bilingual slice prototypes and activates the highest scoring slices within a fixed term budget; each chunk injects at most 10 score-filtered references into the language-model prompt. No separate ASR or source-transcript topic classifier is required.

On a complete four-talk ACL/medicine stream with 848 forced-aligned term occurrences, AutoTerm reaches 0.853 term accuracy, within 0.12 points of a 0.854 known-domain reference. A merged 100k glossary remains competitive at 0.863, but at one million entries accuracy falls to 0.830; AutoTerm also improves BLEU by 1.91 and raw masked BLEU by 1.88 over this 1M baseline. It raises strict time-local prompt precision from 4.4% to 39.3% while reducing references per chunk from 8.79 to 1.12. These results show that a small multi-slice working set preserves useful prompt evidence as the offline catalog grows, without requiring session-time setup. The demo exposes active slices, retrieved terms, and routing evidence in a live interface.

1 Introduction

Simultaneous speech translation (SST) targets technical talks, lectures, meetings, and live media. Prior SST work studies the quality–latency trade-off of prefix translation (Ma et al., 2019) and standardizes streaming evaluation (Ma et al., 2020). Recent omni speech-language models broaden the available modeling options for long-form speech translation (Qwen Team, 2025), but live systems still struggle with term-dense content: named methods, datasets, clinical terms, legal phrases, and financial entities often require specialized translations that are difficult to infer from local acoustic context alone.

Glossaries are a practical way to control terminology, but they are awkward for live use. A user may not know the exact topic before a talk starts, and asking every speaker or viewer to upload a domain glossary or select a domain creates friction. At deployment time, operators register modular, pre-indexed terminology resources for different domains. At session time, users neither upload a glossary nor select a domain: AutoTerm-SST maintains a small active working set from streaming evidence and injects at most 10 retrieved terms into each LLM prompt. This allows new domains to be added without retraining the translation model or collapsing all terminology resources into a single monolithic inventory.

AutoTerm-SST treats terminology control as an active-inventory and ranking problem. RASST uses glossary retrieval to expose out-of-distribution terms to the language model, but the retriever and prompt have finite ranking capacity: as a universal glossary grows, more irrelevant and conflicting references compete for the same evidence budget. AutoTerm adds a cache-like management layer above the unchanged retriever. Semantic similarity admits promising slices, the current slice receives recency-style pinning, and low-priority slices leave the working set when its term budget is full. The motivation is not that million-term retrieval is computationally infeasible, but that a small active set remains more relevant as the offline catalog expands.

The result is an interactive streaming SST workbench rather than a new translation model. The framework handles REST/WebSocket transport, session lifecycle, and agent routing; agents handle model inference, prompting, batching, and retrieval.

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The live agent uses Qwen3-Omni with vLLM continuous batching (Kwon et al., 2023), and the web and Electron interfaces expose retrieved terms, active glossary status, router decisions, and latency alongside translations.

Our contributions are:

- We build AutoTerm-SST, a pluggable streaming SST demo with a live evidence panel for retrieved terminology, active glossary state, router decisions, and latency.
- We introduce zero session-time terminology setup: users neither upload a glossary nor select a domain, while a training-free, budgeted active-slice cache keeps multiple plausible topic glossaries available.
- On a frozen four-talk stream, we compare the working set with nested 100k/1M merged glossaries and a known-domain reference, alongside retrieval scale and 32-session concurrent serving.

2 Related Work

Streaming speech translation. SST systems must translate partial speech while balancing quality and latency. Prefix-to-prefix policies such as wait-k make this trade-off explicit (Ma et al., 2019), attention-based policies refine read/write decisions (Papi et al., 2023, 2024), SimulEval standardizes evaluation (Ma et al., 2020), and large-scale streaming models target direct simultaneous translation (Seamless Communication, 2023; Zhang et al., 2024). Speech language models make this setting more flexible (Qwen Team, 2025; Ouyang et al., 2025), but live deployment still requires session management, incremental transport, and latency-aware prompting. AutoTerm-SST focuses on adapting terminology during a stream; its contribution is a system demonstration rather than a new model architecture.

Terminology-aware translation. Terminology control has long been studied in machine translation, usually through user-provided dictionaries, constrained decoding, or training-time exposure to terminology constraints (Hokamp and Liu, 2017; Dinu et al., 2019); terms and named entities are also a known weakness of speech translation (Gaido et al., 2021). Retrieval-augmented generation offers another way to expose non-parametric memory at generation time (Lewis et al., 2020; Khandelwal et al., 2021). Many of these workflows assume that the domain or terminology is known before translation begins. RASST retrieves glossary evidence for out-of-distribution terms before LLM decoding, but its glossary is fixed before streaming (Luo et al., 2026). AutoTerm-SST keeps the RASST retriever and translation model unchanged and instead manages a budgeted cache of topic slices online. This moves domain selection from user setup into the streaming system and prevents an expanding offline catalog from placing every term in the same ranking competition.

LLM serving and demo systems. Modern LLM serving systems provide continuous batching and efficient generation APIs (Kwon et al., 2023; Zheng et al., 2024), but a streaming speech demo also needs microphone/file input, WebSocket transport, session cleanup, live evidence rendering, and mock-mode testing. Translation Canvas (Dandekar et al., 2024) targets offline translation analysis; AutoTerm-SST provides a live serving layer while keeping model-specific code in swappable agents.

3 System Overview

AutoTerm-SST uses a thin framework with swappable translation agents, while its main system contribution is the adaptive terminology loop inside the live agent (Figure 1). The framework serves the web UI, accepts REST control requests, streams audio over WebSockets, tracks session lifecycle, and routes each session by `agent_type`; it does not own model prompts, retrieval, batching, KV cache, or terminology resources. For chunk t, the agent retrieves from the current budgeted working set before decoding. The generated translation may update that working set for a later chunk, no earlier than $t+1$, but never changes the prompt already used for chunk t.

Agent contract. Agents implement a small contract: `open_session`, `submit_audio`, `on_control`, `close_session`, `describe`, and `health`. The same UI and wire protocol support a deterministic mock agent for development, a legacy InfiniSST scheduler agent (Ouyang et al., 2025), or the AutoTerm-SST live agent built on RASST/Qwen3-Omni (Luo et al., 2026). The boundary is backend-agnostic: the live demo uses in-process vLLM (Kwon et al., 2023), and the same protocol can serve external SGLang-compatible backends (Zheng et al., 2024) without changing

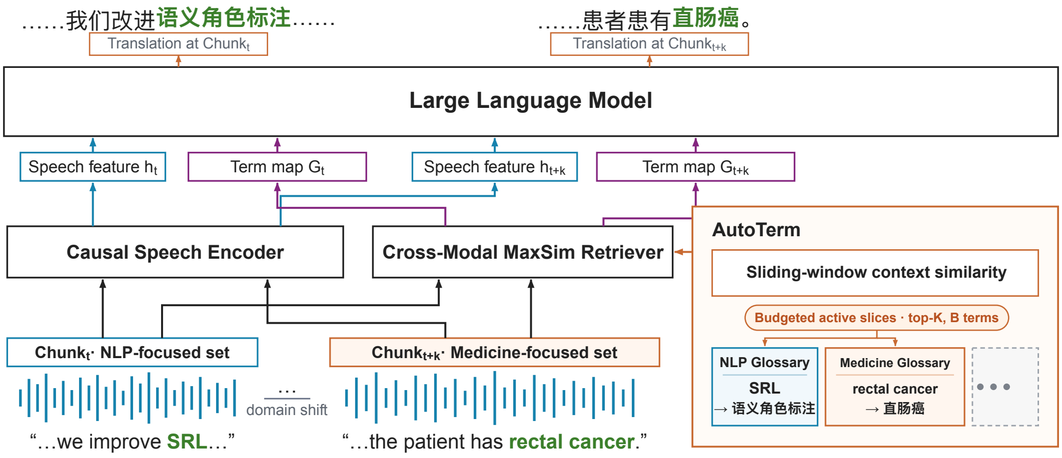

Figure 1: AutoTerm-SST on a stream that moves from NLP to medicine. Each speech chunk is encoded by the RASST backbone, and at most 10 bilingual term pairs retrieved from the *active slices* are injected into the language-model prompt. Sliding-window translation context ranks bilingual slice prototypes under a fixed term budget, so *SRL* is served from the NLP glossary early and *rectal cancer* from the medicine glossary later. The working set may hold several slices; the ellipsis denotes additional registered glossaries.

the terminology memory, router, or UI.

Streaming agent. For each audio chunk, the OmniAgent scores the accumulated target-text window against offline bilingual slice prototypes. It selects the highest-similarity slices that fit the active-term budget, then runs MaxSim retrieval across that working set. The agent filters and jointly reranks candidates, caps prompt references at 10, and lets the PromptBuilder format the `term_map` before calling the in-process vLLM backend. After decoding, y_t is appended to the target history for a later routing tick. The agent emits partial translations and structured metadata, including retrieved terms, retrieval latency, active slices, prompt-reference counts, and router decisions.

Evidence protocol. The default WebSocket output remains plain text for backward compatibility. A JSON mode returns `{type, text, meta}`, where `meta.references` drives the live evidence panel and `meta.topic` and `meta.topic_router` describe the adaptive glossary state. This evidence protocol lets users see which terms were retrieved and whether the router stayed, switched, or deferred.

4 Adaptive Terminology Memory

AutoTerm-SST separates the offline terminology catalog from the small working glossary exposed to the language model. The catalog may contain many topic slices and substantially more entries than should be searched or placed in a streaming prompt at once. Each slice stores term–translation pairs, lightweight topic metadata, and a prebuilt MaxSim index. Exact duplicate pairs may be removed, but the same source term with different domain-dependent translations is retained.

Budgeted semantic routing. Users neither upload a glossary nor choose a domain. At chunk t, the router concatenates the most recent w generated translation chunks into a sliding-window context c_t. Each slice d has an offline bilingual prototype set P_d, consisting of a short description and representative source/target terminology. With a multilingual sentence encoder E, the router computes

$$\bar{p}_d = \operatorname{norm}\left(\frac{1}{|P_d|}\sum_{p\in P_d} E(p)\right), \qquad (1)$$
$$s(d \mid c_t) = \cos(E(c_t), \bar{p}_d)\,.$$

It ranks slices by this similarity and activates the highest-scoring slices that fit a term budget B, up to K slices. This is training-free and allows several plausible topics to remain available: a single wrong top-1 decision need not remove the correct terminology from retrieval. The current best slice is pinned when possible to avoid oscillation at a topic boundary.

Fixed evidence budget. For every chunk, the active indexes share one retrieval candidate budget; the surviving candidates are jointly ranked and at most 10 references enter the language-model prompt. Thus increasing K does not grant AutoTerm more retrieval candidates. In our main comparison, AutoTerm ranks ten 1k slices, uses w=4, K=4, and B=4,000; the 100 retrieval candidates are divided across the active slices before the common score threshold and top-10 prompt cap. The known-domain reference instead searches one 2.5k slice, while the merged baselines search nested 100k- and 1M-entry indexes under the same candidate and prompt budgets (Section 6).

The 1M index is not assumed to be computationally infeasible. The comparison tests a different systems question: as an offline catalog grows, can a compact semantic working set preserve relevant prompt evidence without session-time setup? Broad indexes can still be retained as background resources or rescue pools; AutoTerm controls how much of that resource competes for each streaming prompt.

5 Demo Interface

The demo is designed for three audiences: researchers inspecting terminology errors in streaming SST, lecture or conference participants who do not want to prepare glossaries, and developers testing new speech translation agents behind a stable protocol. The same interface can run in mock mode without a GPU or in live GPU mode with Qwen3-Omni and MaxSim retrieval.

Live controls. Users choose a model agent, language pair, latency multiplier, and terminology mode. In `auto_working` mode, the session starts from a configured default slice without requiring the user to identify the domain, and it may activate other domain slices within its working-set budget; every chunk uses the same top-10 prompt cap. The hosted demo currently exposes lightly curated NLP and medicine glossaries (generic single-word entries removed and session proper nouns added). The stress evaluation uses its own frozen ten-slice registry and unmodified inventories, so the hosted UI resources are not used to construct Table 1. Manual presets and ad-hoc term overrides remain available as advanced options; they affect the prompt but do not bias the router.

Evidence panel. For each partial translation, the UI renders the latest non-empty retrieved references as term-translation pairs with scores and source labels. A status line reports active memory backend, term count, and retrieval/generation latency; the adaptive row adds mode, inferred topic, confidence, active glossaries, switch count, and router action. This makes terminology behavior visible rather than hidden inside the model prompt.

Packaging. The repository includes a mock-friendly launcher for UI/protocol development and a GPU launcher for the live AutoTerm-SST agent built on RASST. Mock mode exercises the same REST/WebSocket flow and JSON evidence protocol, enabling reviewers to test the demo without model weights or GPUs. The live demo additionally uses a resident vLLM engine, a dedicated retriever GPU, and an optional public tunnel.

Availability and licensing. The demo code is released under the MIT license on the GitHub repository. A source checkout is the downloadable package; in mock mode, it starts the same UI and protocol without GPUs. Live Qwen3-Omni mode requires A6000-class GPUs and the external RASST retriever checkpoint; model weights and Wikidata/Wikipedia-derived terminology resources retain their upstream licenses. The hosted demo is reachable through a stable entry page.[1]

6 Evaluation

Setup. We stream four complete talks through one alternating session: ACL 268, medicine 545006, ACL 367, and medicine 606 (5,381.508 seconds in total). The live En→Zh agent runs Qwen3-Omni with vLLM tensor parallelism and the RASST MaxSim retriever on 2×A6000. We compare four terminology policies. *Known-domain-2.5k* activates the matching 2.5k slice from the known talk domain and is a reference policy that requires domain metadata. *AutoTerm-1k×4* requires no session-time setup and activates at most four 1k slices from the recent four-chunk translation context. *Merged-100k* and *Merged-1M* search one universal glossary; 100k is a strict prefix of 1M. All policies use the same 1.92s chunks, 100 total retrieval candidates, score threshold 0.78, top-10 prompt cap, and 2,803 decoder windows.

Metrics. The gold builder yields 1,079 raw annotation rows. We collapse only source aliases with

[1]Live demo: https://luojiaxuan.github.io/autoterm-sst/; screencast video: https://github.com/luojiaxuan/autoterm-sst/blob/main/docs/demo_screencast.mp4.

Figure 2: Web UI during a live En→Zh session, moments after the automatic transition to a medicine-focused working set. The translucent panel renders the JSON evidence stream over the settings card: retrieved term pairs with retrieval scores and their source slice (here `auto:medicine_core_10k`), per-chunk retrieval latency, and the streaming translation with retrieved terminology hits highlighted inline. The latency multiplier was changed live mid-session (status chip).

the same exact audio span and identical target variants, leaving a fixed set of 848 spoken occurrences (615 ACL and 233 medicine); nested terms, different target variants, and domain-dependent translations remain separate. Each occurrence is matched one-to-one to a target variant within a $[-2, +30]$s output window. We also report BLEU, BLEU after masking a curated technical inventory (T-MBLEU), and BLEU after masking the full raw term inventory (R-MBLEU). Strict prompt precision is the fraction of injected references whose source term overlaps a raw gold occurrence in that chunk's local retrieval window; Refs is the mean number of injected references per chunk. The scorer requires byte-identical playlists and exact equality of every persisted decoder window across conditions.

Active-budget comparison. Table 1 shows that AutoTerm matches the known-domain reference: their TERM_ACC differs by only 0.12 points (723 vs. 724 hits), while AutoTerm's BLEU is 0.26 higher. Merged-100k is not a failure case: it has the highest TERM_ACC (732 hits), and its BLEU/R-MBLEU remain within 0.09/0.05 of AutoTerm. However, it already lowers prompt precision to 8.83% and increases references per chunk to 5.00. At 1M, AutoTerm leads by 2.24 TERM_ACC points (723 vs. 704 hits), 1.91 BLEU, 1.90 T-MBLEU, and 1.88 R-MBLEU. Its strict prompt precision is 39.29% versus 4.37%, with 1.12 rather than 8.79 references per chunk. The million-entry baseline is not computationally unusable; under a fixed top-10 evidence budget, irrelevant and conflicting candidates crowd the prompt. AutoTerm keeps a small multi-slice working set while leaving the broad catalog offline.

Figure 3 complements the policy comparison with a longer session-level trace of automatic domain transitions.

Appendix B shows why 1M is the relevant stress point: a controlled ACL sweep remains stable through 100k while retrieval precision falls as distractors grow. Appendix C gives construction details, the full scorecard, and examples of top-10 crowding and domain-dependent translation conflicts.

Concurrent serving. A single live session runs in real time (median first-output latency 0.35s). With vLLM continuous batching on 2×A6000 (TP=2), 32 concurrent sessions stay real-time: 95th-percentile per-segment latency holds near 0.82s (8–32 sessions), below the 1.92s stride. These serving numbers characterize the hosted system and are separate from the terminology-policy comparison.

7 Limitations and Conclusion

AutoTerm-SST is a terminology-aware streaming system, not a new foundation model or an open-world topic classifier. Semantic routing depends on short offline slice descriptions and bilingual prototypes; a novel or ambiguous topic may therefore receive a poor working set. Because the strongest routing evidence is recent generated translation, the top-ranked slice can lag after a domain boundary, although activating several slices reduces the cost of a single wrong top-1 choice. The main comparison covers one En→Zh four-talk stream and ten candidate slices. It is a controlled system result, not evidence that every million-entry glossary or

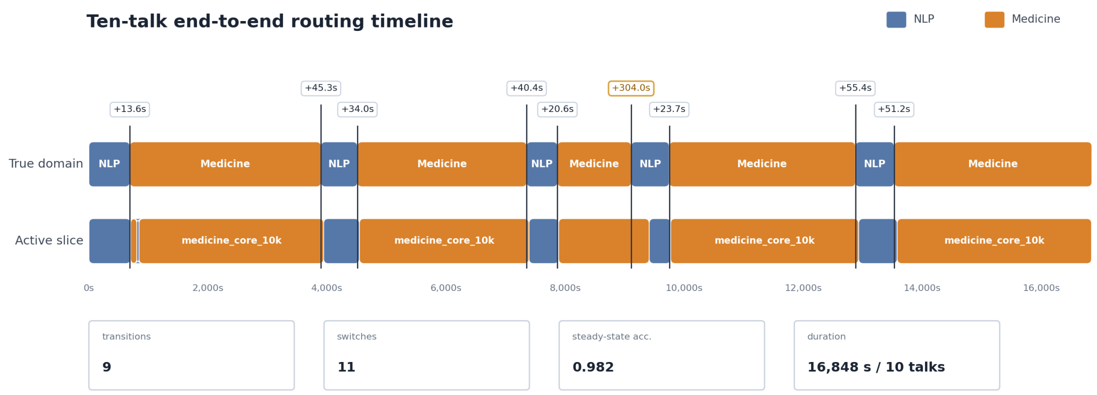

Figure 3: Active-slice routing in a separate, full-length ten-talk alternating stream (five ACL and five medicine talks; 16,848s in one live session). The upper row is the known talk domain and the lower row is the automatically selected slice. All nine domain transitions switch in the correct direction: eight within 13.6–55.4s, and one medicine-to-NLP transition after 304s because of a domain-generic talk opening. The run has one 35s transient wrong-slice excursion and 0.982 steady-state active-domain accuracy outside 30s boundary grace windows.

Setting	Session setup	Active terms	TERM_ACC	BLEU	R-MBLEU	Prompt prec.	Refs
Known-domain-2.5k	Select domain	2.5k	0.854	53.77	50.20	**0.394**	**1.12**
AutoTerm-1k×4	None	≤4k	0.853	**54.03**	**50.46**	0.393	**1.12**
Merged-100k	None	100k	**0.863**	53.94	50.41	0.088	5.00
Merged-1M	None	1M	0.830	52.12	48.59	0.044	8.79

Table 1: Full four-talk comparison under identical decoder windows and retrieval/prompt budgets. TERM_ACC uses the same 848 time-aligned spoken occurrences for every row; the merged inventories are strictly nested.

language pair will degrade by the same amount. In particular, Merged-100k remains competitive in both the controlled ACL sweep and the four-talk comparison, and a 1M MaxSim index is computationally usable; the observed failure is ranking and prompt crowding under a fixed evidence budget. The hosted live backend also requires A6000-class NVIDIA GPUs and external model/retriever artifacts, while mock mode covers UI and protocol testing without them.

We presented AutoTerm-SST, which removes per-session glossary upload and domain selection by routing a compact multi-slice working glossary from streaming context. On the frozen four-talk comparison, this working set improves term and translation quality over a merged 1M glossary while injecting substantially cleaner prompt evidence. The architecture also provides a practical expansion path: new topic slices can be registered offline without forcing every session to search one ever-growing universal glossary.

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A Budgeted Semantic Router Configuration

Table 2 gives the frozen configuration used for the main comparison. The context encoder is used only for slice routing; MaxSim remains the speech–term retriever. Descriptions and bilingual prototype examples are encoded once, averaged within each topic, and normalized. At runtime, the normalized embedding of the accumulated target-text window is compared with these centroids by cosine similarity.

Parameter	Value
Router context	latest 4 translation chunks
Semantic encoder	BAAI/bge-m3
Candidate slices	10
Terms per slice	1,000
Maximum active slices (K)	4
Total active-term budget (B)	4,000
Pin current best slice	yes
Shared retrieval candidates	100
MaxSim threshold	0.78
Prompt references	at most 10

Table 2: Frozen AutoTerm configuration for Table 1. Retrieval candidates share one total budget rather than receiving a separate budget per slice.

The top-1 topic label changes conservatively and can lag a domain boundary. The retrieval policy is less brittle than that label: up to four high-scoring slices remain in the working set, so the correct inventory may be available before it becomes top-1. On the full stream, the three top-1 transitions took 40.4, 43.5, and 82.1 seconds and made no wrong-direction switch. These delays are a limitation of target-context routing, not hidden domain metadata.

B Glossary-Scale Streaming Sweep

Memory	Terms	p50/p95 ms	Refs/chunk
Curated	238	61.3 / 105.3	2.33
Domain	10k	78.5 / 82.1	1.80
Open	100k	64.1 / 82.6	4.89
Stress	1M	78 / 95	9.62

Table 3: Warm per-chunk MaxSim retrieval latency and surviving prompt references by memory size. Cold loading, performed off the streaming path, costs substantially more as the index grows.

To locate the onset of retrieval noise, we also stream five concatenated ACL 60/60 talks (3,442 seconds) through manually selected nested inventories. Every row retains the same 238-entry ACL glossary and only adds distractors. Up to 100k, TERM_ACC and translation quality remain in a narrow band even as prompt precision falls (Table 4). This is why 100k is not presented as a failure case in the main paper; Merged-1M is the capacity stress condition.

Inventory	Terms	Term acc.	BLEU	M-BLEU	Gold@10	Prec@10	Refs
Curated ACL	238	0.972	58.27	53.10	0.972	0.480	1.53
+AI transl.	10,238	0.965	58.72	53.35	0.972	0.234	3.09
+AI broad	10,238	0.972	58.38	52.75	0.979	0.367	2.00
+AI broad	50,238	0.979	58.65	53.24	0.979	0.196	3.64
+AI broad	100,238	0.965	58.83	53.32	0.979	0.137	5.03

Table 4: Controlled ACL capacity sweep. Gold@10 is gold-term coverage within the prompt cap; Prec@10 is retrieval precision; M-BLEU masks technical terms.

C Four-Talk Active-Budget Comparison

Setting	Hits	TERM_ACC	NLP	Med.	Prompt prec.	Refs
Known-domain-2.5k	724	0.854	0.867	**0.820**	**0.394**	**1.12**
AutoTerm-1k×4	723	0.853	0.873	0.798	0.393	**1.12**
Merged-100k	732	**0.863**	**0.885**	0.807	0.088	5.00
Merged-1M	704	0.830	0.867	0.734	0.044	8.79

Table 5: Domain breakdown and prompt evidence for the full stream. Hits and TERM_ACC use 848 spoken occurrences (615 NLP, 233 medicine).

The main comparison preserves domain ambiguity rather than deduplicating it away. We remove only exact source–target pair duplicates; when one source term has different translations in different domains, all mappings remain. The merged baselines combine the target-bearing pool with general distractors to exactly 100,000 and 1,000,000 mappings; the former is a strict prefix of the latter. These extra distractors need not be assigned to artificial topic slices: they model the universal-glossary alternative. AutoTerm uses ten 1k topic slices, and the known-domain reference uses the matching 2.5k NLP or medicine slice. All four policies receive 100 total retrieval candidates, threshold 0.78, and at most 10 prompt references.

We selected slice size, K, and context length on four fixed development windows totaling 570 seconds, then froze the setting before running all four talks. These windows come from the same talks and are configuration-search data, not a held-out evaluation; only the untouched 5,381.508-second run and its 848-occurrence denominator appear in the headline table.